

Life at low temperature: psychrophiles

Pure water freezes at 0 °C under ordinary atmospheric pressure, but that familiar number is not an absolute biological boundary. Natural water contains salts and other dissolved substances, occurs in small pores and films, and may be exposed to different pressures. These conditions can keep a fraction of it liquid below 0 °C. As cooling proceeds, however, the liquid fraction shrinks and its chemistry changes because growing ice leaves most solutes behind.

Microorganisms approaching and crossing the freezing point therefore encounter more than a lower thermometer reading. Chemical reactions and molecular movements slow; membranes pack more tightly; and cells may be confined to increasingly salty liquid channels between ice crystals. The biological challenge is to keep transport, energy conservation, catalysis and gene expression working in this altered physical setting.

Cold-adapted microorganisms solve these connected problems through combinations of membrane remodelling, cold-active enzymes, RNA and protein quality control, compatible solutes and extracellular structures. Each solution addresses a particular consequence of cooling and may carry a cost under warmer conditions. There is no single “cold-adaptation gene” or universal psychrophile design.

1. Life at low temperature: definitions and limits

The term *psychrophile* literally means “cold-loving”, but its scientific use requires more precision. A widely used operational convention defines a psychrophile as an organism with a minimum growth temperature at or below 0 °C, an optimum at or below 15 °C, and a maximum near 20 °C {cite:p}morita1975. Organisms that can grow at low temperature but grow optimally at substantially warmer temperatures are generally described as *psychrotolerant*. The older term *psychrotroph* is still common, especially in food microbiology, where the practical concern is growth during refrigeration.

These categories are useful labels, not natural boundaries. A temperature optimum measured in a nutrient-rich laboratory culture describes the condition that maximises growth rate in that experiment. It does not fully describe fitness in a cold habitat, where nutrients may be scarce, temperature may fluctuate, and salinity or pressure may constrain growth. A psychrotolerant organism with a comparatively warm laboratory optimum may sometimes compete effectively at low temperature, whereas an organism meeting the conventional psychrophile definition may perform poorly under a habitat's actual resource conditions. Psychrophily is therefore best understood as adaptation along a continuum, with cardinal growth temperatures used to state clearly how an organism was classified {cite:p}cavicchioli2016.

Cold-specific records and limits

The general distinction among detection, survival, maintenance, activity, growth and reproduction is developed in Chapter 15, *Extremophiles and life's extremes*. Here the essential point is cold-specific: a temperature record is meaningful only when paired with its biological endpoint and physical conditions.

The strongest controlled growth case retained here is *Planococcus halocryophilus* Or1, a bacterium in the phylum Bacillota isolated from High Arctic permafrost. It grew and divided at –15 °C in a liquid culture medium reported as containing 18% NaCl and 7% glycerol, with an estimated generation time of about 50 days. In a separate sterile-permafrost microcosm, the same strain mineralised radiolabelled acetate at –25 °C

over 210 days {cite:p}mykytczuk2013. The first experiment demonstrates reproduction under high-solute conditions; the second demonstrates low metabolic activity, not division. A natural permafrost community also incorporated radiolabelled acetate into lipids at -20°C , but the active organisms were not identified {cite:p}rivkina2000.

Survival records answer a different question. Many microorganisms that are not psychrophiles can be recovered after storage at -80°C , especially when glycerol or another *cryoprotectant* limits freezing damage. Culture collections also preserve microorganisms in or above liquid nitrogen, near -196°C {cite:p}hubalek2003. Dormant endospores are especially resistant: *Bacillus subtilis* spores retained substantial viability after 360 days at -80°C in a controlled longevity experiment {cite:p}nagler2018. These observations demonstrate survival followed by recovery during warming. They do not demonstrate activity at the storage temperature and should not be used to define the lower temperature for metabolism.

Example	Domain; phylum	Condition tested	Demonstrated endpoint
<i>Planococcus halocryophilus</i> Or1	Bacteria; Bacillota	-15°C , high-solute liquid medium	Growth and cell division
<i>P. halocryophilus</i> Or1	Bacteria; Bacillota	-25°C , sterile-permafrost microcosm	Acetate mineralisation
Natural permafrost community	Mixed Bacteria; taxa unresolved	-20°C , frozen environmental sample	Acetate incorporation into lipids
<i>Bacillus subtilis</i> spores	Bacteria; Bacillota	-80°C for 360 days	Survival and recovery after warming

Liquid water below 0°C

Subzero temperature does not necessarily mean that all environmental water is solid. Dissolved salts lower the freezing point, while solutes rejected from growing ice become concentrated in the remaining liquid. Sea ice contains networks of brine inclusions and channels. Frozen soils and glacial ice may contain pores, veins and thin interfacial films whose physical state and composition depend on temperature, mineral surfaces and solutes. These microscale environments can provide liquid water, but they are chemically demanding: as the liquid fraction shrinks, salinity increases and water becomes less available to cells {cite:p}deming2002.

There is consequently no universal lower temperature for life that can be separated from phase state, solution chemistry, timescale and measurement method. A defensible limit claim must state what biological process was observed, for how long, in which medium or microhabitat, and with which controls.

2. What low temperature does to microbial cells

Temperature influences almost every cellular process, but the effects of cooling are not all versions of “reactions becoming slower”. Cooling changes molecular motion, fluid properties, phase behaviour and macromolecular interactions at the same time. The result is a connected set of challenges rather than a checklist of independent stresses {cite:p}damico2006,demaayer2014.

Slower kinetics, diffusion and transport

Most cellular reactions require molecules to reach configurations that cross an activation barrier. As temperature falls, the proportion of encounters that reach those configurations generally decreases, so reaction rates decline. This is a kinetic effect: it does not mean that every reaction becomes thermodynamically unfavourable. The relationship is also not simply linear. Different reactions can respond differently, rate-limiting steps can change, and phase transitions or acclimation can produce departures from a simple temperature dependence. Chapter 10, *Thermodynamics and kinetics*, develops the quantitative relationship. For the present argument, cooling reduces the rate of many enzyme-catalysed steps while saying nothing by itself about whether the underlying reaction is energetically favourable.

Cooling also increases the viscosity of liquid water. Molecules diffuse more slowly, and transport becomes more difficult, particularly across long distances or through narrow pores and extracellular matrices. Nutrient acquisition, waste removal and encounters between enzymes and substrates can all be affected. In partially frozen environments, the problem is spatially uneven: diffusion through a concentrated brine channel is not equivalent to diffusion through a well-mixed laboratory broth.

Membrane ordering

Biological membranes must remain ordered enough to form a selective barrier but dynamic enough for lipids and proteins to move and change conformation. Cooling packs many lipid molecules more tightly and can push a bilayer towards a more gel-like state. A membrane that becomes too ordered changes its permeability and can restrict the conformational movements of transporters, respiratory complexes and ATP synthase. The consequence is not merely a structural inconvenience. Because cells conserve energy across membranes, a change in membrane physical state can interfere with nutrient uptake, ion homeostasis and the maintenance of electrochemical gradients.

Macromolecular dynamics

Proteins are not rigid objects. Catalysis, ligand binding, folding and assembly depend on movements that occur across a range of timescales. Cooling slows these movements and can reduce catalytic flux even when an enzyme remains correctly folded. Protein folding and the assembly of large complexes can also become slow enough to create bottlenecks. Cold denaturation is possible for some proteins, but it should not be treated as the universal or dominant response of every cellular protein.

Nucleic acids pose a related problem. Lower temperature stabilises many base-paired structures. Stable secondary structures in messenger RNA can obstruct translation initiation or alter RNA processing, while ribosome assembly and turnover of RNA–protein complexes slow. DNA replication, transcription and translation are therefore affected both by chemical kinetics and by the dynamics of the macromolecular structures involved.

Freezing and solute concentration

When extracellular water freezes, most dissolved ions and organic molecules are excluded from the ice lattice. They accumulate in the residual liquid, exposing cells to changing ionic strength, osmotic pressure and water activity. Water can leave a cell as the extracellular solution becomes more concentrated, and growing ice can impose mechanical constraints. Intracellular ice is usually especially damaging because crystals disrupt membranes and macromolecular organisation, but whether ice nucleates inside or outside a cell depends on the cooling regime and the presence of nucleating surfaces {cite:p}deming2002.

The biological problem is therefore coupled: the same fall in temperature that slows catalysis can order the membrane, stabilise RNA structures, reduce the liquid-water fraction and increase solute concentrations around the cell. Adaptation must maintain a workable balance across all of these effects.

3. How cells adapt to cold

Cold adaptation does not reverse the effects of cooling; it changes cellular components so that they continue to function under them. These changes are compromises. A membrane optimised for low temperature may become too permeable when warmed; an enzyme that moves readily enough to catalyse a reaction in the cold may lose stability at higher temperature. The mechanisms below are best understood as matched solutions to particular problems, not traits that every psychrophile must possess.

Membranes and transport

Cells can adjust membrane composition to keep the bilayer within a functional physical range, a response often termed *homeoviscous adaptation*. In many cold-adapted bacteria, changes in fatty-acid unsaturation, chain length, branching or lipid class reduce tight packing at low temperature. These mechanisms are non-exclusive: an organism may use several at once, and the dominant response can change with growth conditions {cite:p}damico2006,demaayer2014.

The familiar statement that psychrophiles “contain more unsaturated fatty acids” captures only one common mechanism. It is not a universal rule. Lipid responses differ among bacterial lineages, and archaeal membranes require separate treatment because their hydrophobic chains are ether-linked to glycerol and can have architectures unlike bacterial fatty-acid bilayers. In the Antarctic methanogenic archaeon *Methanococcoides burtonii*, direct lipid analysis found greater unsaturation of ether-linked phospholipids in cells grown at 4 °C than at 23 °C {cite:p}nichols2004. This demonstrates one archaeal route to temperature-dependent membrane adjustment; it does not define a general rule for all Archaea.

Even within Bacteria, simple composition indices can mislead. In *P. halocryophilus* Or1, the reported ratio of branched to straight-chain fatty acids decreased at the lowest experimental temperatures, contrary to a common expectation for cold adaptation in Gram-positive bacteria {cite:p}mykytczuk2013. The relevant outcome is a membrane that supports selective permeability and protein function, not maximisation of any single lipid category.

Maintaining membrane function supports transport and energy conservation. Transporters must cycle between conformations, respiratory proteins must exchange electrons or ions, and the bilayer must limit uncontrolled leakage. The membrane is therefore a point where molecular adaptation connects directly to metabolic performance. Pigments and carotenoids may modify membrane properties or protect against oxidative and photochemical stress in particular organisms, but pigmentation should not be presented as a general cold-adaptation mechanism without organism-specific evidence.

Proteins, enzymes and metabolic flux

An enzyme can remain folded in the cold yet work too slowly to sustain metabolism. Catalysis requires small changes in molecular shape that bring substrates and catalytic groups into productive arrangements. Cooling slows these *conformational dynamics* and reduces catalytic turnover. A cold-active enzyme can compensate when parts of its structure move productively at lower temperature, often because fewer or weaker stabilising interactions reduce the energetic barrier to those movements {cite:p}feller2003.

This is the core of the *activity–stability–flexibility model*: greater productive flexibility can improve low-temperature catalysis, but weakening stabilising interactions may reduce thermal stability or substrate affinity. The cell gains catalytic speed in the cold at a possible cost under warmer conditions. In the Antarctic bacterium *Arthrobacter* sp. TAD20, the cold-active chitinase ArChb combines efficient low-temperature catalysis with heat lability concentrated in its active-site domain {cite:p}lonhienne2001. Experiments that introduced stabilising interactions towards those of a warmer-adapted homologue improved stability but reduced aspects of cold catalytic performance {cite:p}feller2003.

The simple model is a starting point, not a universal law. Productive movement may be local to an active-site loop rather than a property of the whole protein. Different enzyme families alter electrostatics, solvent access, conformational sampling and stability in different combinations, and some cold-active enzymes retain high overall stability. Amino-acid trends—such as fewer salt bridges or changes in proline, glycine or arginine content—are therefore possible routes, not a sequence recipe from which cold adaptation can be inferred.

Cells must also produce and maintain proteins under conditions where folding and assembly are slow. Molecular chaperones can prevent unproductive interactions, folding catalysts can accelerate rate-limiting rearrangements, and proteases can remove damaged or persistently misfolded products. These systems support cold growth, but their presence does not diagnose psychrophily: mesophiles also deploy chaperones and proteases during transient cold shock. The scientific question is whether a system has a demonstrated role in sustained low-temperature function, not whether its gene appears in a genome.

At the scale of metabolism, cold-active catalysis helps maintain flux but does not erase energetic limitations. Slow substrate supply, membrane transport and maintenance costs can still constrain growth. A purified enzyme's activity is therefore evidence about one molecular step, not a direct measure of the organism's growth rate in nature.

Nucleic acids and gene expression

Cooling stabilises RNA base pairing and slows the formation and turnover of RNA–protein complexes. Structures that are harmless or short-lived at a warmer temperature can persist long enough to obstruct translation initiation, ribosome assembly or RNA degradation. Cells address this problem with proteins that bind RNA, prevent unproductive structures or remodel existing complexes.

DEAD-box proteins are an important example. These ATP-dependent RNA remodelling proteins can disrupt short RNA duplexes and reorganise ribonucleoprotein complexes. In *M. burtonii*, a DEAD-box RNA helicase was shown to be regulated by low temperature and the purified protein could unwind double-stranded RNA {cite:p}lim2000. The result supports a mechanistic connection between cold and RNA remodelling. It does not, by itself, quantify the protein's contribution to organismal fitness, because the study did not establish that contribution through a genetic loss-of-function experiment.

Cold-shock proteins provide another source of confusion. In a mesophile, rapid cooling can trigger a transient response that stabilises translation and protects nucleic acids while the cell acclimates. Over longer periods, membrane composition, enzyme abundance and other physiological states may adjust. A psychrophile adapted over evolutionary time may express some comparable systems constitutively or regulate them differently. *Cold shock*, *acclimation* and *long-term adaptation* consequently describe different timescales and must not be treated as synonyms.

Genomics and transcriptomics can reveal candidate mechanisms. A gene enriched in cold-environment genomes may have been selected for cold function, but it may instead reflect shared ancestry, another environmental variable or an incompletely annotated role. A transcript that increases after cooling establishes regulation under the experimental conditions; it does not prove that the product is adaptive. Stronger inference requires a chain of evidence: biochemical function, genetic perturbation, a relevant cellular phenotype and, ideally, improved performance under the environmental condition. Omics data are powerful for generating that chain's starting hypotheses, not for replacing it.

Ice, compatible solutes and extracellular microenvironments

Ice-interacting systems act through several distinct mechanisms. An *ice-binding protein* adsorbs to an ice surface and may inhibit crystal growth or recrystallisation, change crystal shape, or mediate adhesion. An *ice-nucleating protein* promotes ice formation at a warmer subzero temperature than would otherwise occur, the opposite physical action from inhibiting ice growth. Whether either physical effect benefits the producing organism is system-dependent and requires direct evidence {cite:p}dolev2016.

The Antarctic bacterium *Marinomonas primoryensis* demonstrates an unexpected use of ice binding. Direct observations in a microfluidic system showed motile cells attaching to ice; antibodies directed against the ice-binding region blocked this attachment {cite:p}bardolev2016. Structural work showed that the ice-binding region forms part of a very large, multidomain surface adhesin that can connect the bacterium with ice and diatoms {cite:p}guo2017. In this system, the strongest demonstrated cell-level role is adhesion and positioning near an ice surface, not secretion of a soluble antifreeze that prevents the surrounding lake from freezing.

Compatible solutes address a different consequence of freezing. Cells accumulate these small organic molecules to balance osmotic stress while disturbing enzymes and other cellular chemistry relatively little; particular solutes can also stabilise proteins or membranes. Their synthesis or uptake can therefore support activity in concentrated brines, although the molecules and effective concentrations differ among organisms. A compound used for laboratory cryopreservation is not automatically a natural microbial adaptation.

Extracellular polymeric substances (EPS) extend adaptation beyond the cytoplasmic membrane. These heterogeneous matrices can promote adhesion, retain water and solutes, protect extracellular enzymes and alter a cell's immediate physical environment. The marine psychrophile *Colwellia psychrerythraea* 34H (Bacteria, Pseudomonadota) illustrates both genuine psychrophily—its growth optimum is about 8 °C—and extracellular interaction with ice. Native EPS was retained preferentially relative to salts in laboratory-grown saline ice {cite:p}methe2005,marx2009,ewert2011. Heating removed the preferential retention, but the active heat-labile component was not identified. The experiment therefore demonstrates an interaction between an EPS preparation and saline ice, not a universal polysaccharide antifreeze.

Ice-binding proteins, compatible solutes and EPS can thus influence the liquid microenvironment in complementary ways. Their physiological value must be established at the appropriate level. Crystal-shaping in vitro is evidence of a physical interaction. Demonstrating that the interaction improves survival, activity or reproduction requires an experiment on cells under relevant conditions.

4. Cold never acts alone

Natural cold habitats often impose interacting environmental constraints because temperature changes the effects of other variables. The clearest example is the interaction between cold and salinity. When water

freezes, salts are excluded into the residual liquid. Continued cooling decreases the liquid fraction and can raise solute concentrations, lower water activity and alter ionic ratios. A cell in sea ice or frozen soil is therefore not exposed to “cold plus salt” as two independent stresses: freezing itself connects them {cite:p}deming2002.

In sea ice and frozen soil, cells must manage both membrane ordering and the loss of water to increasingly concentrated brine. Compatible solutes and selective ion transport can help balance this osmotic and ionic challenge. Chapter 21, *Halophiles*, develops the underlying water-activity and osmoadaptation mechanisms in full.

The deep ocean couples cold with high hydrostatic pressure. Both variables affect membrane organisation, protein conformations, transport and reaction rates, and their combined influence need not equal the sum of effects measured separately {cite:p}deming2002. Cells must maintain molecular assemblies that remain sufficiently dynamic in the cold without being destabilised by pressure. Chapter 22, *Piezophiles*, explains pressure adaptation, while Chapter 26, *Oceans*, places these constraints along the water column and seabed.

Radiation, desiccation and limited resource supply can also modify cold adaptation, but the general principles are not repeated here. Chapter 16, *Polyextremophiles*, explains how interacting environmental parameters can produce non-additive biological effects.

5. Ecology and evolution

Cold-adapted microorganisms operate across an enormous fraction of the planet. About 90% of ocean water is at 5 °C or below, mainly because the deep ocean remains cold at all latitudes {cite:p}rodrigues2008. In these waters and sediments, microbial heterotrophs return organic carbon to inorganic forms and regenerate nutrients. In a -0.7 °C deep-sea sediment experiment, benthic bacteria dominated the processing of added algal and faecal-pellet carbon, and algal carbon was mineralised more than three times faster than the lower-quality faecal material {cite:p}mayor2012. The result connects cold-active metabolism to the residence time and burial of carbon at the seabed rather than merely showing that cells are present there.

Permafrost provides a second planetary-scale example. Arctic permafrost regions contain nearly 1,700 billion tonnes of frozen and thawing carbon. When ice-rich ground thaws, previously protected organic matter becomes accessible to microbial decomposition. Aerobic conditions favour CO₂ production, whereas waterlogged, oxygen-limited settings can increase the contribution of microbial CH₄ production {cite:p}miner2022. The size and timing of these fluxes remain uncertain, but the mechanism is specific: thaw changes liquid-water access, redox conditions and substrate availability, allowing microbial metabolism to mobilise a vast carbon reservoir.

Microorganisms also alter the physical behaviour of glacier and snow surfaces. *Cryoconite* is a dark aggregate of mineral particles, organic matter and microbial cells; solar heating can melt the surrounding ice and create water-filled cryoconite holes where cyanobacteria and heterotrophs cycle carbon and nutrients. On snow, blooms of pigmented microbial eukaryotes produce red or pink fields. The chlorophyte *Sanguina nivaloides* (Eukarya, Chlorophyta) is a representative red-snow alga whose carotenoid-rich resting cells tolerate intense radiation and freeze-thaw conditions. Across 16 Arctic glaciers and permanent snowfields, red-snow blooms were estimated to reduce seasonal snow albedo by about 13%, increasing absorbed solar energy and promoting melt {cite:p}lutz2016,prochazkova2019. This *bioalbedo* effect is locally important, but its total contribution still depends on bloom coverage, weather and the duration of the melt season.

These examples also show why habitat temperature alone cannot predict ecological success. A brine channel, deep sediment and melting snow may all be cold, yet differ in pressure, salinity, light, substrate supply and disturbance. Growth rate matters, but so do substrate affinity, maintenance cost, resistance to fluctuations and interactions with other organisms. Detailed process budgets belong in the chapters on oceans, permafrost and ice, but the connection to psychrophily is direct: cold-adapted metabolism controls whether carbon is transformed, retained or released under the prevailing physical conditions.

Psychrophily occurs among Bacteria, Archaea and microbial Eukarya. It is therefore a physiological and ecological strategy, not a branch on the tree of life. Similar functional outcomes have arisen through different molecular routes: bacterial and archaeal membranes can both be adjusted with temperature even though their lipid chemistry differs; different enzyme families can achieve cold activity through different structural changes; and RNA remodelling can be regulated in lineage-specific ways. This distribution is consistent with repeated evolutionary solutions to the same physical problems {cite:p}lauro2013.

Care is needed when moving from functional similarity to evolutionary history. Showing that two lineages both contain cold-active enzymes does not establish that a particular molecular trait evolved independently in each. Claims about independent origins require explicit phylogenetic evidence. **Cold imposes recurring physical problems, and evolution has produced multiple solutions rather than one universal psychrophile programme.**

6. Research frontiers and broader significance

The lower limit for sustained reproduction remains unresolved. Record temperatures depend on the liquid phase, solute chemistry, incubation duration and sensitivity of the assay. Experiments lasting months can reveal processes invisible in shorter studies, but they also make contamination, background reactions and the distinction between maintenance and growth especially important. Rather than seeking a single number detached from context, future work must define which process stops, under which chemical conditions, and on what timescale.

The activity–stability–flexibility model is another productive frontier. It explains strong experimental examples, but its generality across enzyme families and lineages remains uncertain. Combining kinetics, structural biology and measurements of local conformational dynamics can distinguish mechanisms that are hidden by the broad statement that a protein is “more flexible”.

Functional validation is the major challenge for omics-based studies. Comparative genomics can identify genes associated with cold habitats, and transcriptomics can identify responses to cooling. The next steps are perturbation and phenotype: removing or modifying the candidate, measuring the molecular function, and testing whether the change alters growth or activity under controlled cold conditions. Experiments must also separate temperature from correlated factors such as salinity, growth rate and nutrient state.

Research on polyextremophily increasingly asks how variables interact rather than simply which stress-response genes are present. Membrane composition, compatible-solute pools and extracellular matrices may contribute to several constraints simultaneously. Factorial experiments and measurements under environmentally realistic phase conditions are needed to identify whether combined effects are additive, antagonistic or synergistic.

Cold terrestrial habitats also serve as constrained analogues in astrobiology. Ice shells and cold planetary surfaces may contain saline liquid phases, so studies of terrestrial brine channels help define how temperature, salt composition, water activity and energy supply interact. Psychrophiles show that Earth life

can function in some cold brines; they do not demonstrate that a particular environment on Mars, Europa or Enceladus is habitable, still less that it contains life {cite:p}renno2021. Chapter 48, *Life beyond Earth: astrobiology*, develops the planetary evidence and the limits of analogue reasoning.

Mechanism-led applications

Cold-active enzymes can catalyse reactions at temperatures where warmer-adapted homologues perform poorly, and heat lability can permit easy inactivation after a process. These properties motivate low-temperature food, detergent and molecular-biology processes, although a promising enzyme assay does not establish commercial use or whole-process energy savings. Chapter 45, *Extremophiles' contributions to society*, compares validated applications and process-level costs.

Ice-interacting systems also inspire technological ideas, but their mechanisms must be identified. In one recent laboratory study, extracellular supernatants from three cold-grown bacterial cultures changed freezing and thawing measurements in a frost-susceptible soil system {cite:p}rahman2024. The active molecules were not identified, and the result does not demonstrate field-scale frost control or show that antifreeze or cold-shock proteins caused the effect. Its value is as a frontier example of a measurable phenomenon whose molecular basis and practical relevance remain open.

7. Key concepts

- Psychrophile categories are operational. Cardinal growth temperatures help classify organisms but do not fully describe ecological fitness.
- Cooling slows reaction kinetics, increases viscosity, orders membranes, stabilises some macromolecular structures and changes water from a uniform solvent into ice plus solute-concentrated liquid microhabitats.
- Adaptations address particular consequences: membrane remodelling supports transport and energy conservation; altered enzyme dynamics support catalytic flux; RNA remodelling supports gene expression; and compatible solutes or extracellular systems modify freezing and osmotic conditions.
- Growth at -15°C , activity at -25°C and recovery after cryogenic storage are different cold records and cannot be substituted for one another.
- Natural cold habitats often impose interacting environmental constraints. Freezing couples cold to salinity and water availability, while deep environments couple cold with pressure.
- Psychrophily occurs across Bacteria, Archaea and microbial Eukarya, with multiple molecular solutions rather than one universal programme.

Further reading

- Morita's foundational review explains the cardinal-temperature convention and the history of psychrophile terminology {cite:p}morita1975.
- Cavicchioli's commentary provides the clearest short critique of rigid temperature categories and laboratory optima as measures of environmental adaptation {cite:p}cavicchioli2016.
- D'Amico and colleagues offer a concise mechanistic synthesis of the principal cellular challenges and responses {cite:p}damico2006.
- De Maayer and colleagues connect physiology with genomic hypotheses while illustrating why candidate mechanisms still require functional validation {cite:p}demaayer2014.
- Mykytczuk and colleagues report the *P. halocryophilus* experiments that distinguish growth at -15°C from activity at -25°C {cite:p}mykytczuk2013.

- Feller and Gerday develop the biochemical basis and limitations of the activity–stability–flexibility model {cite:p}feller2003.
- Deming links subzero liquid microhabitats with salinity, pressure and polar microbial ecology {cite:p}deming2002.
- Dolev and colleagues review the distinct structures and functions of ice-binding proteins {cite:p}dolev2016.